Detection of Tea Leaf Blight in Low-Resolution UAV Remote Sensing Images

Gensheng Hu, *Member, IEEE*, Ruohan Ye, Mingzhu Wan, Wenxia Bao, Yan Zhang, and Weihui Zeng

***Abstract*— The accurate detection of tea leaf blight (TLB) from low-resolution (LR) images of tea plants is a challenging task. The small size, high density, and blurred edge details of diseased tea leaves in the LR images from remote sensing with an unmanned aerial vehicle (UAV), as well as the complex backgrounds in the images and the similarity between the color and texture features of TLB spots and the background, cause state-of-the-art methods to give low detection accuracy. This study proposes a two-stage network that incorporates a super-resolution (SR) network RFBDB-GAN with a lightweight detection network (LWDNet) for the accurate detection of TLB in LR UAV images. RFBDB-GAN is applied to recover the details of UAV images. A multiscale two-stage degradation is introduced to simulate the degradation of UAV images caused by the use of various shooting heights to reduce the solution space of RFBDB-GAN. RFB dense residual block (RFBDB) and spectral normalization (SN) term are added to the generator and discriminator, respectively, to recover the texture details and make the TLB spots more obvious. An SIoU loss and an enhanced lightweight PANNet (ELPNet) are employed in LWDNet to achieve accurate TLB detection using a model with a very small size. Experimental results show that RFBDB-GAN yields noticeable improvements over other SR methods, and the model size of LWDNet is 716 kB or 5% of the size of the baseline YOLOv5s. Compared with other state-of-the-art methods, the proposed method has a higher detection accuracy for TLB in LR UAV remote sensing images.**

***Index Terms*— Deep learning, lightweight network, object detection, super-resolution (SR), tea diseases, unmanned aerial vehicle (UAV) remote sensing.**

I. Introduction

TEA has been cultivated in China for more than 2000 years and has served as a vital conduit for trade and cultural interactions with other nations. The Chinese government vigorously promotes the trade in tea and places enormous importance on this business. For a vast number of tea producers in China, planting and selling tea is the primary route to financial success. In 2022, China's tea plantation covered 3.3 million hectares, yielded 3.35 million tons of tea, and generated an estimated 300 billion yuan in total revenue, meaning that the tea industry has grown to be a major economic force. However, tea leaves are often infected with diseases during the growth process, which can seriously affect the quality and yield of tea and cause huge economic losses to tea farmers [1]. Thus, the real-time monitoring of tea growth is critical.

Tea leaf blight (TLB) is one of the most frequent diseases of tea plants, appearing in all of the country's major tea-planting districts, and is generally caused by an infection of the anthracnose pathogen in camellias. Early symptoms of TLB include yellow-brown and water-soaked disease spots, and in the middle stages, the leaves will form uneven patches of light brown and gray. A severe TLB infection will result in damage to leaves and the appearance of dead shoots, weakening the tree [2].

Currently, the majority of TLB detection is done visually in the field by experienced tea growers or plant protection experts. The subjective and environmental influences that frequently affect traditional manual detection methods mean that these are arduous and time-consuming detection procedures, with inconsistent detection findings and high error rates. In addition, since most tea gardens are spread out over a large region and located in remote mountainous places that are difficult for experts to reach, detecting TLB disease in a tea garden may require a great deal of manpower and material resources. As remote sensing technology has advanced, remote sensing tools and methods have provided innovative new methods of crop disease detection. Several studies have adopted satellite, airplane, on-ground equipment, and unmanned aerial vehicle (UAV) platforms to obtain images for the detection of crop diseases. However, due to the weather conditions and observation intervals involved, aircraft and satellite technologies are subject to many constraints in terms of spatial/temporal resolutions and observation capabilities [3]. Furthermore, these solutions are highly expensive. On-ground devices have limited coverage and take a long time to cover wide regions. In contrast, UAV remote sensing is more affordable, can be operated quickly, and is simpler to use than other data collection techniques. Many researchers have combined UAV remote sensing with machine learning methods in the domain of disease and pest detection for crops,

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and this approach has replaced manual detection methods as an effective means of long-term agricultural management.

Machine learning detection methods for crop diseases and pests can be roughly divided into two categories: traditional machine learning and deep learning methods. In order to distinguish between foregrounds and backgrounds for the purposes of detection, traditional machine learning relies on manually produced features such as color, texture, and edge gradients in conjunction with classifiers such as support vector machine (SVM) or random forest (RF). Marin et al. [4] designed a machine learning algorithm to detect coffee plant rust from UAV images, in which vegetation indices, such as green ratio vegetation index (GRVI) and green difference vegetation index (GDVI), were extracted and decision tree models, such as logistic model tree (LMT), functional trees (FT), and random tree (RT), were used to classify the vegetation indices to achieve detection of coffee rust. Siebring et al. [5] proposed a target-based image analysis method for the detection of potato disease from ultrahigh-resolution UAV images. Their method used a large-scale mean shift to segment potato plants in the image, and an RF model was then applied to detect disease in the segmented image. Han et al. [6] used a UAV to collect visible and multispectral images of corn and adopted SMOTE and ENN methods to process the unbalanced dataset. They extracted the relevant features from remote sensing images and constructed an XG boost model to detect lodging corn. Zhou et al. [7] employed a UAV to obtain grape canopy data and then compared and analyzed the partial least squared regression (PLSR), support vector regression (SVR), RF regression (RFR), and extreme learning regression (ELR), with the aim of monitoring grape diseases. Lei et al. [8] combined multispectral and visible light data collected by a UAV to obtain multiple vegetation indices and created multiple models based on algorithms such as decision tree, naive Bayesian, and SVM to detect yellow leaf disease in betel nuts. Similarly, Albetis et al. [9] extracted multiple spectral features from multispectral images of four selected vineyards in southwest France using a UAV and applied SVM algorithms to detect and classify the severity of grape chlorosis in affected vines. Yu et al. [10] took UAV hyperspectral images of pine trees and used a vegetation index combined with an RF algorithm to detect pine wilt disease. Although the use of traditional machine learning techniques for monitoring plant diseases can save effort and energy compared to manual methods, they rely on the manual design of appropriate features and the feature representation ability is insufficient, making it difficult to fully extract the discriminative feature information of the object images. The effectiveness of traditional machine learning techniques is not optimal, especially when the background of the object is complicated and the image has low resolution (LR).

Deep learning methods have stronger feature description and model generalization capabilities than traditional machine learning methods, which can significantly reduce the cost of manual intervention. In the area of agricultural information technology, the combination of deep learning methods with UAV remote sensing technologies has opened up new research possibilities for the detection of crop diseases and pests. Ishengoma et al. [11] used UAV remote sensing technology to capture maize leaves infected by the autumn night moth and compared the performance of fine-tuned VGG16, VGG19, and other models in terms of detecting infected areas. The results showed that the fine-tuned VGG16 yielded significantly improved detection accuracy. Oliveira et al. [12] described a method that combined U-Net and PSPNet networks to detect nematodes in high-resolution (HR) RGB UAV images of coffee plants. Kerkech et al. [13] proposed a multimodality method of detecting grape disease from UAV images based on optimized image matching and segmentation with the SegNet network. They registered and integrated visible and infrared UAV remote sensing images of grape trees and used the SegNet network to detect the presence of grape mold. Kerkech et al. [14] constructed a model for the detection of grape disease using a convolutional neural network (CNN), which combined different color spaces and vegetation indices and extracted HSV, LAB, and YUV color and texture features with a CNN. The detection results showed that a combination of the YUV color space and the excess green minus excess red index (ExGR) vegetation index yielded the best outcome. Chen et al. [15] adopted a YOLOv3 network to detect litchi stink bug from UAV images and reported that the mAP for pest detection reached 90%. Yu et al. [16] used a UAV equipped with a multispectral camera to obtain images of pine trees suffering from wilt disease and applied a two-stage object detection algorithm that combined a Fast R-CNN with YOLOv4 to detect disease. Shi et al. [17] developed a new end-to-end deep learning model called CropdocNet for the accurate detection of potato late blight from UAV hyperspectral images. They considered the potential disease-specific changes in reflectance radiation caused by the diversity of the canopy structures and introduced multiple capsule layers to process the spectral features extracted from HSIs. However, in the methods described above, the UAV had a low flight height, the resolution of the captured images was high, and the area of diseased crops covered by each image was very small, resulting in a low efficiency of detection.

Tea is grown more densely than other crops, and the leaves are crowded and piled. The majority of current disease detection methods for tea rely on data collected from close-range photography using high-definition cameras or mobile devices. Hossain et al. [18] obtained HR tea images with an SLR camera and then used SVM to analyze 11 features of these images to achieve the detection and recognition of tea diseases. Recently, Jayapal and Poruran [19] proposed a detection algorithm based on a deep hash integrated self-encoder that could detect spots in HR images of tea captured with mobile phones. Hu et al. [20] expanded a tea dataset collected with a camera via C-DCGAN and then used a VGG16 model to detect tea diseases. Although the methods mentioned above have achieved relatively good results for the detection of tea disease from HR images, the resolution of UAV images of tea plants is typically low, the size of the diseased leaves is small, and the edge details of the spots are blurred. In addition, the images have complicated backgrounds, and the color and texture of the objects are

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similar to those of the backgrounds, which is particularly detrimental in terms of the detection of diseased spots. This means that existing methods are not suitable for the detection of TLB from LR UAV remote sensing images. Choudhury et al. [21] captured images of tea with a UAV and adopted the YOLOv5 network to detect areas of the plants that were infected by the red spider mite (RSM). They could only perform preliminary detection of infected areas, and the detection accuracy of diseased spots on a single leaf was insufficient. Zhao et al. [22] collected HR images of tea using a UAV with a low flight altitude of 3 m and applied Faster R-CNN to detect diseased tea plants. Due to the use of HR images, the mAP reached 76.07%.

Numerous researchers have considered super-resolution (SR) tasks to improve the resolution of images. Cui et al. [23] proposed a new network called RestoreDet for detecting objects from degraded LR images. Their approach gave better detection results for larger objects. Dai et al. [24] conducted research and analysis on the usefulness of SR images in other visual applications and showed that the detection and segmentation outcomes improved after the introduction of the SR method to visual tasks such as edge detection and semantic segmentation. Wang et al. [25] analyzed the effectiveness of SR methods for the very LR image classification tasks. A DSRL model proposed by Wang et al. [26] included additional SR blocks to improve the semantic segmentation performance. Shermeyer and Van Etten [27] explored the application of SR and evaluated the effectiveness of SR preprocessing steps when applied to object detection from satellite images. Haris et al. [28] jointly optimized single-shot multibox detector (SSD) detection loss [29] with an SR network [30] to improve object detection performance.

The SR research mentioned above has mainly focused on objects at large scales or with regular shapes, whereas the size, shape, and background are all irregular when spots are detected from LR UAV images of tea, meaning that existing SR methods are not particularly effective. At the same time, due to the intensive planting of tea trees, the leaves are crowded and stacked, which poses a certain challenge in terms of the detection of disease spots. This study proposes a two-stage network to accurately detect small objects related to TLB in LR UAV tea images. In the first stage, we employ an RFB dense residual block (RFBDB)-GAN network to recover the LR images, and in the second, we use lightweight detection network (LWDNet) for the precise detection of TLB.

The contributions of this study are given as follows.

1) A detection method for TLB in LR UAV remote sensing images is proposed in which an area of 500 m^2 can be covered in a single image.
2) An RFBDB-GAN SR network with multiscale receptive field dense residual blocks is proposed for recovering the details of LR UAV remote sensing images.
3) A multiscale two-stage degradation strategy is proposed to allow us to synthesize a training dataset for the RFBDB-GAN network. This approach resolves issues related to the weak generalization performance and insufficient robustness of deep learning models caused by a lack of training samples.

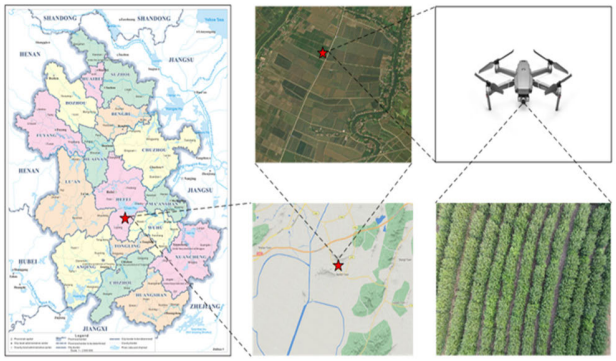


Fig. 1. Data acquisition site.

4) An efficient LWDNet model is proposed to achieve accurate TLB detection. The size of the model is 716 kB, corresponding to only 5% of the size of the baseline YOLOv5s.

II. Materials

A. Data Acquisition

The data on tea plants used in this study were collected at the Bazhen Town tea garden, which is located in the southwest of Hefei, Anhui, China, and is bordered to the west by Wuhu city and to the north by Chaohu Lake. Fig. 1 shows the data acquisition site. The data acquisition system included a UAV with a 48-MP visible light sensor.

There were 128 high spatial resolution images (flight altitude: 2–4 m) used for training RFBDB-GAN and LWDNet. The images captured within this height range have high spatial resolution and clear details, and the cropped training images show obvious regions of disease. In addition, 16 low spatial resolution images (flight altitude: 12–14 m) were used for validation/testing. The training dataset was augmented using sliding windows, resulting in 100 images (800 × 600) from each 8000 × 6000 original image. Some overexposed and blurred images were removed, giving 996 high spatial resolution training images. Fig. 2 shows cropped tea images at different heights: Fig. 2(a) at 4-m altitude and Fig. 2(b) at 12-m altitude. To compare details, we randomly chose a patch (red box) and displayed it in the second row of Fig. 2(a) and (b).

B. Data Preprocessing

From Fig. 2, as the UAV altitude increases, the image spatial resolution decreases, and the TLB spots in the images become smaller, blurrier, and harder to distinguish, thus complicating TLB detection. To address this, we processed LR tea images with an SR network (RFBDB-GAN) to recover leaf and spot details, followed by TLB detection using LWDNet. Also, to satisfy the training requirements of RFBDB-GAN and LWDNet, the training dataset was preprocessed in two distinct steps. Table I shows the number of images in the training dataset before and after preprocessing.

1) Data Preprocessing for RFBDB-GAN: As an SR network, RFBDB-GAN aims to recover SR images from LR tea plants images with unknown and complicated degradation. Existing SR methods usually use classical degradation [31]

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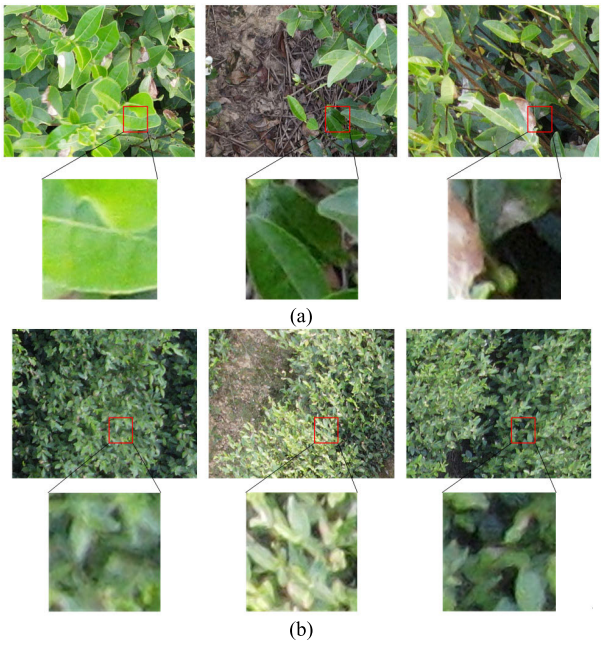


Fig. 2. Cropped tea images with different spatial resolutions and enlargements of small areas. UAV images taken at (a) 4-m altitude and (b) 12-m altitude.

TABLE I
NUMBERS OF IMAGES IN THE TRAINING SET BEFORE AND AFTER PREPROCESSING

Network	Before	After
RFBDB-GAN	996	3996
LWDNet	996	3930

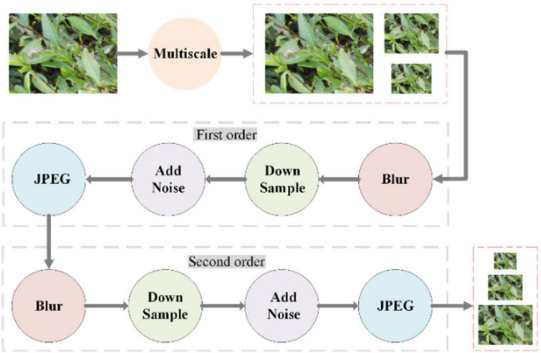


Fig. 3. Sketch map of the proposed multiscale two-stage degradation method.

to synthesize the LR input. Equation (1) is a mathematical expression of classical degradation model

$$x = D(y) = [(y \otimes k) \downarrow_s + n]_{\text{JPEG}} \tag{1}$$

where D represents the degradation process, y is the HR image, k is the fuzzy kernel, $\otimes$ is the convolution operation, s is the downsampling factor, n represents noise, JPEG represents image compression processing, and x is the degraded LR image.

However, as shown in the SR ablation experiments in Section IV, the classical degradation cannot effectively simulate the degradation process due to different UAV shooting heights, resulting in insufficient detailed information in SR images. To enhance the detailed information of the images, a multiscale two-stage degradation method is proposed. As shown in Fig. 3, the proposed degradation

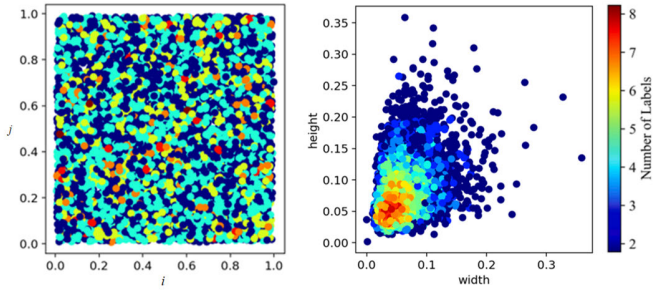


Fig. 4. Normalized spots' distribution map and normalized spots' size map.

method first performs multiscale downsampling on the HR images and then applies two classic degradation operations. In order to adapt to data obtained at different heights, a random scaling factor is introduced to replace the downsampling factor in (1). The degradation method in this study is expressed as follows:

$$x = D^2(y) = (D_2 \circ D_1)(y). \tag{2}$$

2) Data Preprocessing for LWDNet: The leaves in the image are thick and shaded from each other. To avoid labeling errors, we labeled only TLB disease spots, not entire diseased leaves. The locations and sizes of spots were analyzed as a whole to obtain the relative location distribution and size of the labeled spots. Fig. 4 shows normalized spots' distribution and size maps, indicating the presence of numerous small objects in the dataset. To avoid model overfitting due to the small objects and limited samples, LWDNet adopts a mosaic data augmentation method. In addition, to simulate the complex environment of UAV photography in natural scenes, LWDNet also applies flip, color jittering, and noise addition as data augmentation methods.

III. METHODOLOGY

A. Overview

A two-stage strategy for accurately detecting TLB in LR UAV tea images is designed in this study. In the first stage, the RFBDB-GAN is adopted to increase the resolution of LR tea images. In the second stage, the LWDNet is used for detecting TLB.

The methodology mainly consists of five steps.

Step 1 (Image Preprocessing and Training Set Construction): The UAV tea images, which were captured at low altitudes, are cropped. For the RFBDB-GAN, the multiscale two-stage degradation is performed to obtain LR images. For the LWDNet, noise addition, flipping, color jittering, and mosaic operations are employed to expand the training set.

Step 2 (The Training of the RFBDB-GAN): The LR images are input to the generator to generate SR images. The SR images are then compared with the original images by the discriminator to determine the similarity, and the judgment results are fed back to the generator to adjust the generator's parameters. The pseudocode of training RFBDB-GAN is referred to as Algorithm 1.

Step 3 (RFBDB-GAN Test): The LR test images that were captured at 12–14 m are super-resolved by the trained RFBDB-GAN. The generated SR images are then adopted as the

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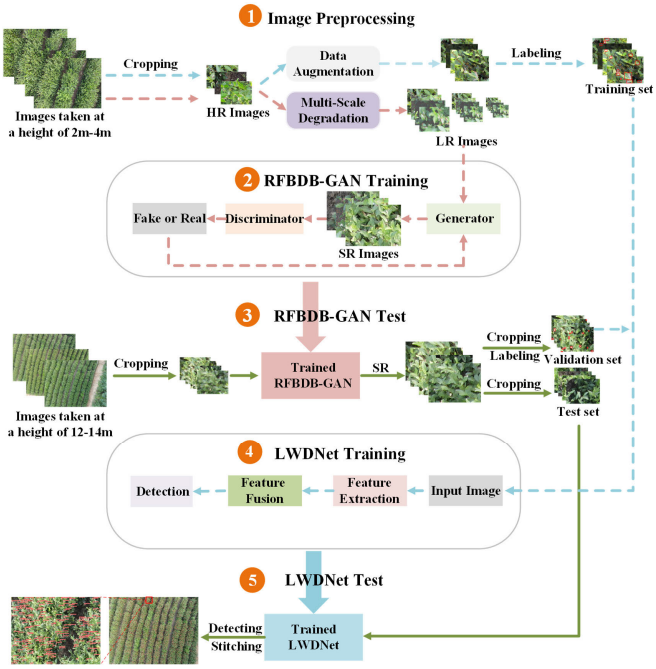


Fig. 5. Detection process of TLB in LR UAV images.

validation and testing set for the detection network. The TLB in SR images are labeled with the guidance of experts.

Step 4 (The Training of the LWDNet): The training and validation set are resized to 640 × 640. The image features are extracted, the feature maps are fused through the enhanced lightweight PANNet (ELPNet) structure, and the TLB is detected by the detection part of LWDNet. The pseudocode of training LWDNet is referred to as Algorithm 2.

Step 5 (LWDNet Test): The SR test set is tested by the trained LWDNet and the detection results are stitched into complete large images.

The detection process is shown in Fig. 5, where the red dashed line represents the training process of RFBDB-GAN, the blue dashed line represents the training process of LWDNet, and the green solid line represents the test process.

B. RFBDB-GAN SR Network

SR methods based on CNN networks mainly focus on minimizing the mean square error (MSE) for the reconstruction, and generated SR images often have a high peak signal-to-noise ratio (PSNR); However, they usually lack high-frequency details, which cannot meet the authenticity of HR and are not visually satisfactory [31]. SR methods based on GAN networks can recover images with high perceptual quality and abundant detail. A GAN network was therefore adopted in this study to recover the details from LR UAV images of tea plants. Due to the irregular shapes in the background and the different texture details in images, existing SR methods based on GAN networks yield poor results when applied to a single LR image of tea. For this reason, we propose an RFBDB-GAN SR network with multiscale receptive field dense residual blocks. The structure of the network is shown in Fig. 6, and the improvement is due to the use of Real-ESRGAN [33]. The generator of

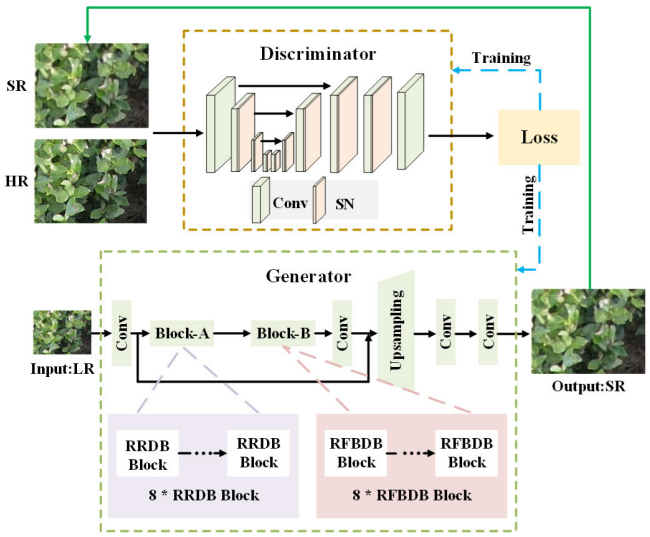


Fig. 6. Structure of the proposed RFBDB-GAN network.

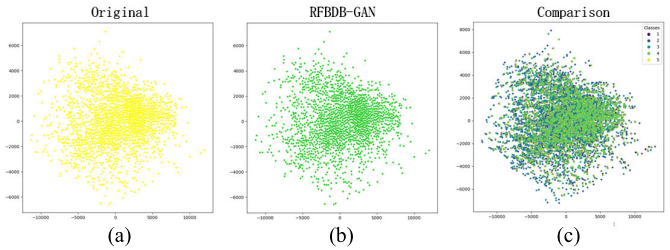


Fig. 7. Distributions of LR data and SR data. (a) Distribution of LR data. (b) Distribution of SR data recovered using RFBDB-GAN. (c) Comparison of data distributions for images recovered by different SR methods.

RFBDB-GAN is a dense residual structure cascaded with RRDB and RFBDB, and the discriminator adopts a UNet with spectral normalization (SN) [34] regularization terms.

We used the L1 loss, perception loss, and GAN loss to jointly train the RFBDB-GAN network. The loss function is shown as follows:

$$\begin{aligned} L_G &= L_{\text{percep}} + \lambda L_G^{\text{Ra}} + \eta L_1 \\ L_1 &= E_{x_i} \| G(x_i) - y \|_1 \\ L_G^{\text{Ra}} &= -E_{x_r}[\log(1 - D_{\text{Ra}}(x_r, x_f))] - E_{x_f}[\log D_{\text{Ra}}(x_f, x_r))] \end{aligned} \tag{3}$$

where L_1 is the 1-norm distance that is applied to evaluate the loss between the HR image (y) and the SR image [$G(x_i)$]; L_{percep} is the perception loss [35]; L_G^{Ra} is the GAN loss; x_f represents the SR image; x_r represents the input HR image; D_{Ra} is the relativistic average discriminator [36]; $E_{x_f}[\cdot]$ represents the operation of taking the average over all the SR image data in the mini-batch; and λ and η are coefficients used to balance the different loss terms, in which $\lambda = 0.1$ and $\eta = 1$ in this study.

In order to verify whether the proposed RFBDB-GAN method generated too much false information, the t-SNE [37] visualization tool was used to draw the data distributions of the original images and the SR images and to compare these with the data distributions after SR using various methods such as ESRGAN [38] and RCAN [39]. Fig. 7(a) shows the distribution of the original LR data, Fig. 7(b) shows the SR

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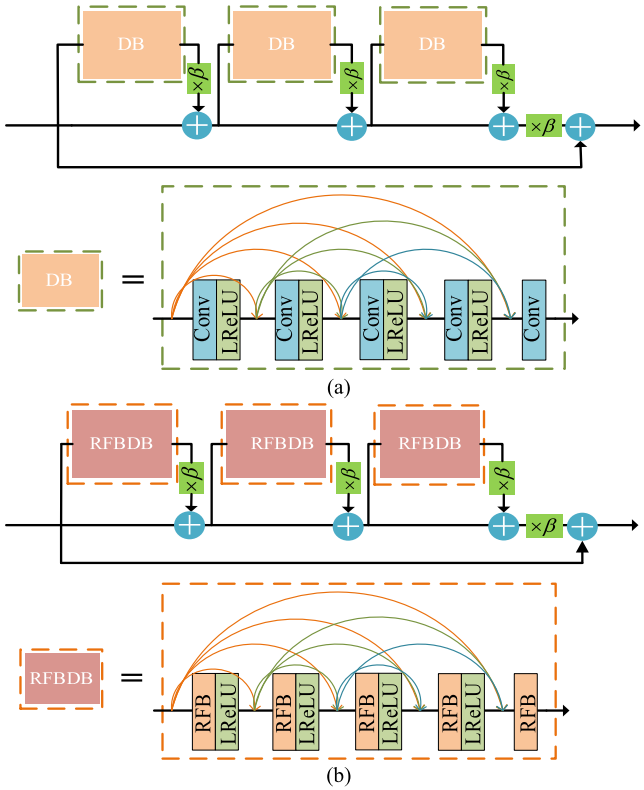


Fig. 8. Structures of (a) RRDB and (b) RFBDB blocks.

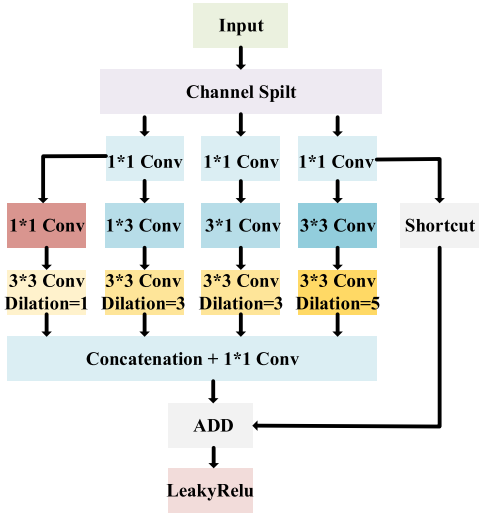


Fig. 9. Structure of the improved RFB.

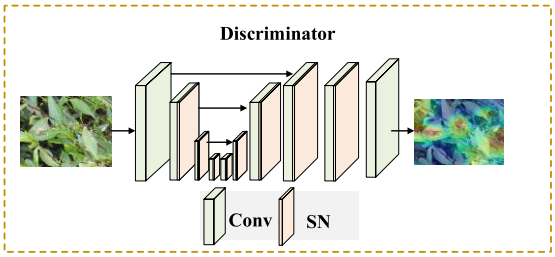


Fig. 10. Structure of the discriminator.

data distribution after RFBDB-GAN recovery, and Fig. 7(c) shows a comparison of the SR data distributions recovered by several other SR methods. Categories 1–4 in Fig. 7(c) represent the SR data recovered by ESRGAN, Real-ESRGAN, RCAN, and RFBDB-GAN, respectively, while category 5 represents the original data. From the visualization results, it can be seen that compared to methods such as ESRGAN, the distribution of the RFBDB-GAN super-resolved data is basically consistent with that of the original data. This indicates that the use of RFBFB-GAN can better preserve the original information contained in the data, without introducing false information.

1) Generator: Here, we describe Block-A and Block-B in the generator that is used to create the details of the image. Block-A consists of eight RRDB blocks that are connected in series, each containing three DB modules with a dense residual structure, as shown in Fig. 8(a). Block-B is composed of eight RFBDB blocks, each containing three RFBDB modules. Unlike in the RRDB block, we replace the convolutional structure in the DB module with an improved RFB [40] structure in the RFBDB module. As shown in Fig. 8(b), the output of each layer in the RFBDB module is connected to the subsequent input layers, thereby promoting information exchange and avoiding redundant feature learning. Parameter β in Fig. 8 is a scale factor for the network and was set to 0.2 for this study based on the experience.

For this study, a UAV was used to obtain images of tea plants at different flight altitudes. Due to the uneven terrain of the tea plantation, the relative heights of the UAV and the tea plants vary even at the same flight altitude. In order to fully utilize the multiscale information contained in the data, an improved RFB structure is adopted in the RFBDB module. This structure can enhance the deep features learned from lightweight CNN models. More specifically, the RFB uses different sizes of receptive field to pool multiple branches of different kernels, applies extended convolution layers to control the eccentricity, and reconstructs them to obtain the final output. The purpose of using the RFB structure in the RFBDB module here is to retain high-dimensional information to restore the details of the SR images. We therefore modify the RFB structure accordingly, removing all batch normalization (BN) layers from the RFB, and use a LeakyReLU rather than a ReLU activation function at the output, although each branch still applies a ReLU activation function. The structure of the improved RFB is shown in Fig. 9. It can extract features of different scales from the feature map, i.e., can extract both coarse and fine features from the input image.

2) Discriminator: Our approach employs a UNet with SN terms as the discriminator for the RFBDB-GAN, as shown in Fig. 10. The function of SN is to ensure that the discriminator satisfies Lipschitz continuity. The underlying idea is to decompose the weight matrix output by each layer of the discriminator into SVD, limit the maximum singular value to one, and divide it by the maximum singular value after each update, which normalizes the weight matrix in the discriminator into the corresponding spectral norms. The network layer after the SN satisfies Lipschitz continuity, meaning that during the optimization process, the parameter changes are more stable and gradient explosions are less likely to occur. The main structure of UNet consists of two parts:

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a convolutional encoding unit and a convolutional decoding unit. In the encoding unit, a 2×2 Maxpooling operation is performed to downsample the input image, and the size of the image is reduced by a factor of 2. After each downsampling stage, the number of features is doubled to compensate for the loss of resolution. The decoding unit consists of upsampling blocks, which upsample the input using convolution. In our framework, UNet outputs realistic values for each pixel and provides the detailed pixel-by-pixel feedback to the generator. Experimental results indicate that when the UNet model is used alone as the discriminator for an RFBDB-GAN network, it cannot restore detailed textures and may even generate some unnatural textures, which increases the instability of the training process. In addition to the first and last convolutional layers, we add SN after each convolutional layer of the discriminator to improve the ability of the RFBDB-GAN network to recover the details of the images. Our experimental results show that the SN term can improve the recovered texture while stabilizing the training dynamics.

3) SR Algorithm for UAV Tea Images: Algorithm 1 gives the pseudocode of training RFBDB-GAN network for SR of UAV tea disease image.

Algorithm 1 An Algorithm for RFBDB-GAN Training

Input: HR-images set $\Omega^H = \{x_1^H, \ldots, x_N^H\}$, number of iterations K, batch size m.

Output: Trained generator G.

1: Degrade the HR-images using Eq.(2), then obtain the paired LR-images dataset $\Omega^L = \{x_1^L, \ldots, x_N^L\}$.
2: Train the generator G individually with L_1 loss and then initialize G.
3: **for** $k = 1$ to K **do**
4: **Train D**;
5: Draw Ω^f from G, discriminate whether the images is true or false.
6: Update the D via gradient-based optimizers based on Eq.(3).
7: **Train G**;
8: Sample batch of m LR images samples from Ω^L
9: Generate m fake images $\Omega^f = \{x_1^f, \ldots, x_m^f\}$ and put these images in discriminator.
10: Update the G via gradient-based optimizers based on Eq.(3).
11: **end for**

C. LWDNet Detection Network

When the RFBDB-GAN has performed the SR process on the LR UAV images, a detection network is then applied to detect TLB from the images. Our framework uses YOLOv5s as the baseline network, and an LWDNet detection network is constructed. The model size of LWDNet is only 5% of the baseline network. The structure of the LWDNet detection network is shown in Fig. 11 and consists of three parts: feature extraction, feature fusion, and detection parts. In order to improve the detection accuracy of the network, the SIoU [41] loss function is used.

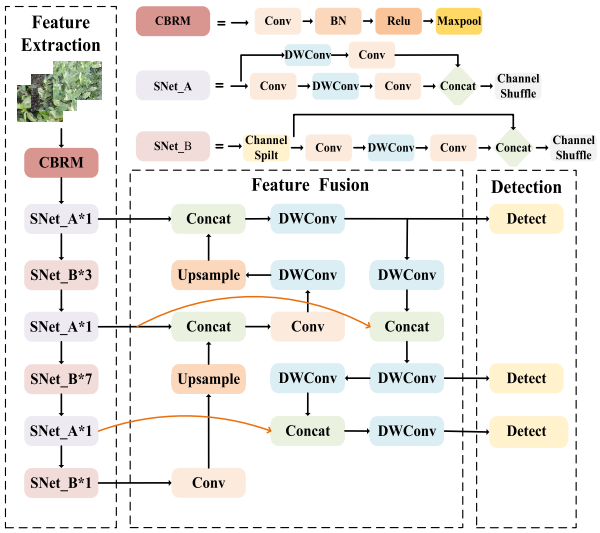


Fig. 11. Structure of LWDNet.

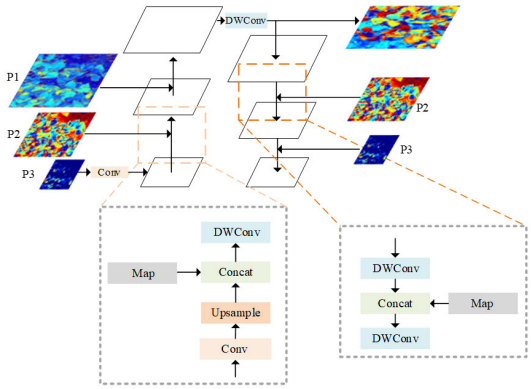


Fig. 12. Structure of the ELPNet feature fusion module.

1) Feature Extraction: The healthy parts of the tea in the image appear green, while the diseased parts appear gray-brown, indicating a large number of duplicate pixels in the image. In order to reduce the computational complexity, reduce the number of model parameters, and avoid computational redundancy, the input image is first passed to the CBRM module, which mainly applies Maxpooling. Then, a structure based on an alternating cascade of SNet_A and SNet_B is adopted as the feature extraction part of the network. SNet_A and SNet_B are shown in Fig. 11. In SNet, 5×5 DWConv is used instead of 3×3 DWConv in ShuffleNetV2 [42], and a LeakyReLU activation function is used rather than a ReLU activation function. The SNet_A structure outputs twice as many channels as input channels, but the output feature maps are half the size of the input feature maps, enabling downsampling of the feature map. The number of channels output by SNet_B is consistent with the number of input channels, and the size of the output feature maps is consistent with that of the input feature maps. Due to the use of a feature pyramid structure in the feature fusion part, which includes two upsampling processes, this study uses three SNet_A modules for downsampling in the feature extraction part.

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2) Feature Fusion: In order to ensure that the network is lightweight and to improve the detection accuracy of LWDNet, we designed an ELPNet feature pyramid structure for the feature fusion part, which integrates features from the shallow, medium, and deep levels, as shown in Fig. 12. P1–P3 are the feature maps output by the feature extraction part. In contrast to the feature extraction structure of YOLOv5s, in order to fully utilize the image information obtained from the feature extraction part, the feature fusion module of the LWDNet detection network fuses the middle-level (P2) and deep-level (P3) feature maps from the feature extraction part multiple times. In the bottom-up process, two upsampling operations are performed to concatenate the shallow and middle feature maps from the feature extraction part. The feature map undergoes convolution before upsampling, and DWConv is used to adjust the number of channels after the Concat operation to ensure consistent input and output channels during the bottom-up process. In the top-down process, the middle and deep feature maps extracted from the feature extraction part are fused separately. To reduce the number of parameters, only DWConv is used in the top-down process. Before the Concat operation, the size of the DWConv kernel is 3×3 with a step size of two, which is used to perform $2\times$ downsampling of the feature map. After the Concat operation, the DWConv kernel size is 3×3 with a step size of one, which is used to adjust the number of channels to ensure consistent input and output channels.

3) Loss Function: LWDNet is formed from a large number of lightweight modules, resulting in a significant reduction in the size of the network from 14.4 MB for the baseline network to 0.7 MB. A decrease in the number of model parameters usually comes at the cost of sacrificing accuracy. To avoid an excessive decline in accuracy, the SIoU loss is employed to replace the original CIoU [43] loss. Loss functions related to IoU [44], such as GIoU [45], DIoU [43], and CIoU, do not consider the direction between the real box and the prediction box, resulting in slow convergence. The SIoU loss function used in this study considers the vector angle between the real box and the prediction box and redefines the related loss function to include the angle loss, distance loss, shape loss, and IoU loss. The SIoU loss function is defined as follows:

$$\text{Loss}_{\text{SIoU}} = 1 - \text{IoU} + \frac{\Delta + \Omega}{2} \tag{4}$$

where Δ represents the distance loss and Ω represents the shape loss. The shape loss is defined as

$$\Omega = \sum_{t=w,h} (1 - e^{-w_t})^{\theta} = (1 - e^{-w_w})^{\theta} + (1 - e^{-w_h})^{\theta} \tag{5}$$

$$w_w = \frac{|w - w^{\text{gt}}|}{\max(w, w^{\text{gt}})}, \quad w_h = \frac{|h - h^{\text{gt}}|}{\max(h, h^{\text{gt}})} \tag{6}$$

where (w, h) and $(w^{\text{gt}}, h^{\text{gt}})$ represent the width and height of the prediction box and the real box, respectively, and θ controls the degree of attention to the shape loss. In order to avoid focusing too narrowly on the shape loss and to reduce the movement of the prediction box, the range of the θ parameter is specified as in [2] and [6].

The distance loss is defined based on the angle loss. The angle loss is calculated as follows:

$$\begin{aligned} \Lambda &= 1 - 2 * \sin^2\left(\arcsin\left(\frac{c_h}{\sigma}\right) - \frac{\pi}{4}\right) \\ &= \cos\left(2 * \left(\arcsin\left(\frac{c_h}{\sigma}\right) - \frac{\pi}{4}\right)\right) \end{aligned} \tag{7}$$

$$\frac{c_h}{\sigma} = \sin(\alpha) \tag{8}$$

$$\sigma = \sqrt{\left(b_{c_x}^{\text{gt}} - b_{c_x}\right)^2 + \left(b_{c_y}^{\text{gt}} - b_{c_y}\right)^2} \tag{9}$$

$$c_h = \max\left(b_{c_y}^{\text{gt}}, b_{c_y}\right) - \min\left(b_{c_y}^{\text{gt}}, b_{c_y}\right) \tag{10}$$

where c_h is the difference in height between the center points of the real box and the predicted box, σ is the distance between the center points of the real box and the predicted box, $\arcsin(c_h/\sigma)$ is equal to the angle α, $(b_{c_x}^{\text{gt}}, b_{c_y}^{\text{gt}})$ are the center coordinates of the true box, and (b_{c_x}, b_{c_y}) are the center coordinates of the predicted box.

After calculating the angle loss, the distance loss can be obtained as

$$\begin{aligned} &\Delta = \sum_{t=x,y} (1 - e^{-\gamma p_t}) = 2 - e^{-\gamma p_x} - e^{-\gamma p_y} \\ &p_x = \left(\frac{b_{c_x}^{\text{gt}} - b_{c_x}}{c_w}\right)^2, p_y = \left(\frac{b_{c_y}^{\text{gt}} - b_{c_y}}{c_h}\right)^2, \quad \gamma = 2 - \Lambda \end{aligned} \tag{11}$$

where (c_w, c_h) are the width and height of the minimum bounding rectangle of the real box and the prediction box, respectively.

4) Detection Algorithm: Algorithm 2 gives the pseudocode for the training process of the LWDNet network used to detect TLB from UAV images.

Algorithm 2 An Algorithm for LWDNet Training

Input: Tea images dataset $X = \{(x_1, y_1), \ldots, (x_N, y_N)\}$, number of epochs K, batch size m.

Output: Trained LWDNet.

```
Split X into training set X_t and validation set X_v randomly.
Initialize parameters of LWDNet θ_0.
for k = 1 to K do
    Train LWDNet with X_t.
    for x_vi ∈ x_v do
        Predict the labels of {x_vi, y_pi}.
        Compare the predicted values with ground truth values based on Eq.(4).
        Update θ_0 to θ_k.
    end for
end for
```

IV. Experiments and Discussion

A. Experimental Metrics

1) Evaluation Metrics for SR Results: The quantitative evaluation metrics of image quality used in this study included the PSNR, structural similarity (SSIM) [46], NIQE [47], and

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TABLE II
CONFUSION MATRIX

Confusion matrix		Actual class	
		TLB	Background
Predicted class	TLB	TP	FP
	Background	FN	TN

TABLE III
EXPERIMENTAL SETTINGS

Component	Specification
CPU	Intel Core i7-9700
GPU	Nvidia GeForce RTX 2080Ti
Operating system	Ubuntu 20.0
Accelerated environment	CUDA11.0 CUDNN8.0
Development environment	PyCharm 2020
Library	PyTorch 1.7.0+cu101

LPIPS [48]. The higher the values of PSNR and SSIM, the better the reconstruction effect. PSNR is an image quality evaluation metric based on error sensitivity. However, following the development of GAN networks, researchers have found that high PSNR or SSIM values may not necessarily indicate better reconstruction quality, as the textural details of the image may not necessarily conform to the visual expectations of the human eye. Researchers have therefore proposed NIQE and LPIPS as metrics of reconstruction quality for an image. The smaller the values of NIQE and LPIPS, the better the reconstruction effect.

2) Evaluation Metrics for Detection Results: Four common metrics for object detection, recall, precision, mAP, and F1, are adopted here. These quantitative evaluation metrics are calculated based on a confusion matrix. An example of a confusion matrix is shown in Table II, where true positive (TP) refers to samples that are accurately detected by the network as TLB, false positive (FP) represents samples incorrectly detected by the network as TLB, false negative (FN) represents the samples incorrectly predicted by the network as background samples, and true negative (TN) refers to samples correctly detected by the network as background samples.

Precision, recall, mAP, and F1 are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{12}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{13}$$

$$\text{mAP} = \frac{\sum_{i=1}^{n} \text{AP}_i}{n} \tag{14}$$

$$\text{F1} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{15}$$

B. Implementation Details

The GPU, operating system, and deep learning framework used in the experiment were an NVIDIA GeForce RTX 2080Ti, Ubuntu, and PyTorch, respectively. The specific experimental details are shown in Table III. The learning rate of RFBDB-GAN was $1e^{-4}$, and the parameter values for Adam optimization were $\beta_1 = 0.9$ and $\beta_2 = 0.99$. The initial learning rate of LWDNet was $1e^{-2}$, using the SGD optimizer with a momentum matrix of 0.937 and a weight decay of 0.0005.

Images taken at heights of 2–4 m and the corresponding degraded images were combined to form data pairs, which were used to train RFBDB-GAN. The training process was divided into two stages: in the first stage, the PSNR-guided generator was trained for 1000k iterations with the L1 loss, while in the second, a combination of the L1 loss, perceptual loss, and GAN loss, with weights {1,1,0.1}, was used to train the generator and discriminator. LWDNet was trained with the SIoU loss until the network converged. The training data for LWDNet consisted of images taken at heights of 2–4 m, whereas the validation and testing data consisted of images taken at a height of 12–14 m and recovered by RFBDB-GAN.

Due to cost and physical limitations on the experimental equipment, higher resolution images taken at a height of 12–14 m were unable to be obtained, and this study therefore had to register images taken at a height of 2 m and used them as references for the ground truth. Fig. 13. shows an example of the registration of an image taken at a height of 2 m with an image taken at a height of 12 m. At the data collection stage, this study used a UAV with a fixed flight height of 2 m and captured images that represented complete coverage of the tea plantation. When annotating the LR UAV images, images taken at 2 m were registered with images taken at 12–14 m, and the former were used as the ground-truth reference for the latter. Under the guidance of experts from the National Engineering Research Center for Agro-Ecological Big Data Analysis and Application, this study then used the detail information from the 2-m images as a reference to carry out the annotation of TLB in the 12–14 m images.



(a) (b)

Fig. 13. Registration of (a) image taken at a height of 2 m and (b) image taken at a height of 12 m.

C. Experiments on the SR Method

In this section, the experimental results from SOTA SR methods are presented, and then, ablation experiments on RFBDB-GAN are described.

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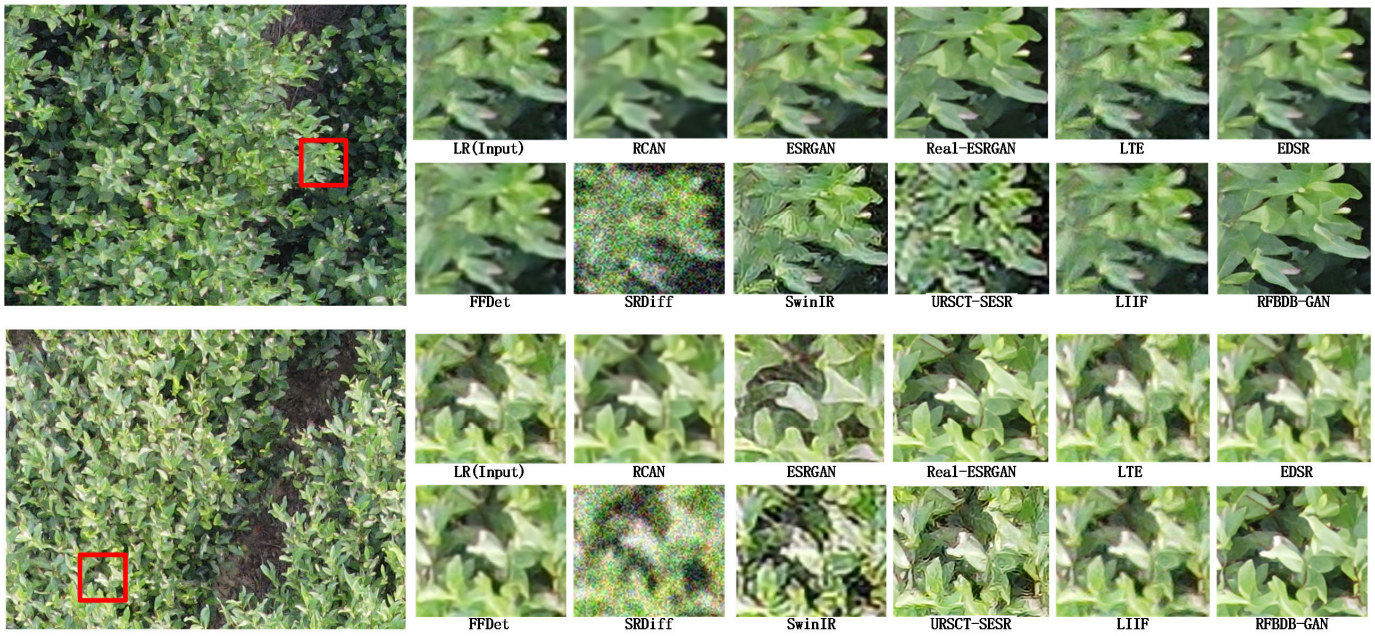


Fig. 14. Comparison of results of different SR networks.

TABLE IV
QUANTITATIVE EVALUATION OF RESULTS FROM DIFFERENT SR NETWORKS

Method	Magnification	LPIPS↓	NIQE↓	SSIM↑	PSNR↑
RCAN	4	0.282	21.618	0.929	30.41
EDSR	4	0.541	14.692	0.802	27.60
FFDNet	2	0.411	16.342	0.916	32.57
LTE	4	0.402	16.618	0.837	35.50
LIIF	4	0.418	16.021	0.884	36.60
SwinIR	4	0.635	15.163	0.542	17.44
URSCT-SESR	2	0.476	14.228	0.853	28.88
SRDiff	4	0.932	26.435	0.160	14.06
ESRGAN	4	0.254	14.231	0.821	25.72
Real-ESRGAN	4	0.221	13.595	0.886	26.03
RFBDB-GAN	4	0.217	13.067	0.887	26.38

1) Results From Different SR Networks: RFBDB-GAN was compared with GAN-, CNN-, diffusion-, and transformer-based SR methods, and the results are shown in Table IV and Fig. 14. The SOTA SR methods were trained on the same training set and were compared on a test set of images collected at a height of 12–14 m by a UAV. CNN-based SR methods (EDSR [49], FFDNet [50], RCAN, LTE [51], and LIIF [52]) typically focus on minimizing the MSE for the reconstruction. The estimated images often have high values of PSNR but often lack high-frequency details, are unable to meet the facticity of HR, and are not satisfactory from a visual perspective. Transformer-based SR methods (SwinIR [53] and URSCT-SESR [54]) have strong local information model ability. However, the receptive field of the plain self-attention is limited within a small area. It prevents the models from utilizing the texture or pattern of neighbor areas to recover degraded pixels, producing the distorted images [55]. Diffusion-based SR method (SRDiff [56]) is not suitable for our data, as there is a lot of noise in the output. GAN-based SR methods (ESRGAN and Real-SERGAN) can generate images with high perceptual quality and more image details. However, the ESRGAN generates some artifacts, which are extremely detrimental in terms of the detection task. Both Real-ESRGAN and the proposed RFBDB-GAN can generate SR images that match human perception, but in the images recovered by RFBDB-GAN, the edge information is more abundant and the edge details of the tea leaves are more obvious.

2) Ablation Experiment for RFBDB-GAN: To improve the reconstruction ability of the model, we developed the RFBDB-GAN network and added SN terms to the discriminator to avoid artifacts in the recovered images. A multiscale two-stage degradation method was also introduced to construct the training data. To demonstrate the effectiveness of the degradation method and SN terms, ablation experiments were conducted.

The results of the ablation experiment of classical one stage and the proposed degradation methods are shown in Fig. 15 and Table V. From Fig. 15, it can be seen that the one-stage degradation method generates artifacts in the SR image

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TABLE V
QUANTITATIVE EVALUATION RESULTS OF SR ABLATION EXPERIMENTS

Multi-scale two-stage degradation	Classical one-stage degradation	SN	LPIPS↓	NIQE↓	SSIM↑	PSNR↑
—	✓	—	0.266	15.224	0.821	23.883
✓	—	—	0.223	14.555	0.886	25.879
—	✓	✓	0.245	14.464	0.865	24.906
✓	—	✓	0.217	13.067	0.887	26.383



(a) (b) (c)

Fig. 15. Results of ablation experiments using different degradation methods. (a) Original image. (b) Classical one-stage degradation. (c) Multiscale two-stage degradation.



(a) (b) (c)

Fig. 16. Results of ablation experiments with or without SN. (a) Original image. (b) Without SN. (c) With SN.

and loses the information on the diseased areas, whereas the multiscale two-stage degradation process yields better effects. As shown in Table V, the results from the multiscale two-stage degradation method can be seen to be superior to those from the classical one-stage degradation method.

The role of the SN term was discussed in Section III-B. Adding SN terms to a discriminator can improve the local detail. Fig. 16 shows the examples of ablation experiment results with and without the SR term in RFBDB-GAN. It can be seen that the model without SN often generates artifacts, and the detailed information is not sufficiently rich. When the SN term is added to the discriminator, the RFBDB-GAN network has a stronger ability to recover the details of the tea images. As shown in Table V, the SN term gives an improvement in various evaluation metrics, and in particular, when the multiscale two-stage degradation method is combined with the SN term, the best SR results are found.

D. Experiments on the Detection Method

In this section, we compare the performance of YOLOv4 [57], YOLOv5, YOLOv7 [58], Faster R-CNN [59], SSD [60], DiffusionDet [61], Deformable-DETR [62], and the proposed LWDNet. This section is divided into three stages.

1) The proposed LWDNet is applied to detect TLB from the SR images generated by different SR networks.
2) Different detection networks are compared in terms of detecting TLB from the SR images generated by the proposed RFBDB-GAN.
3) Ablation experiments on LWDNet are carried out.

1) Detection Results From the SR Images Generated by Different SR Networks: In this section, we use LWDNet to detect TLB from the 4× SR images generated by the SR methods mentioned above (except FFDNet and URSCT-SESR, since they only support SR with a magnification of 2). Fig. 17 compares the detection results from the LWDNet for SR images, where bicubic refers to the ×4 upsampling of the original LR tea images using the bicubic method. In the SR experiments described above, the SOTA methods and the proposed RFBDB-GAN were considered. In visual terms, the RFBDB-GAN network yielded the best results for the recovery of LR UAV tea images. The proposed RFBDB-GAN employs the RFB structure in the generator and also introduces the multiscale degradation method, which are beneficial to the recovery of detailed information and hence to the detection of TLB.

Fig. 18 shows the examples of the detection results from LWDNet when applied to SR images generated by different methods. The resolution of the images in Fig. 18 is 3200 × 2400. It can be seen that the SR images generated by RCAN, SwinIR, SRDiff, and so on are prone to missed and false detection. The SR images generated by Real-ESRGAN are also prone to false detection. This occurs because the SR images highlight the bright spots in the image caused by light, leading detection network to mistakenly detect them as disease spots. The detection accuracy is higher when the proposed RFBDB-GAN is used, as the network is able to detect very small spots in the image, and the false detection of bright spots is reduced.

2) Detection Results From Different Detection Networks: In this section, the training set for the network is constructed using HR images (taken at a low altitude of 2–4 m), and the test set is constructed from SR images (taken at a height of 12–14 m) generated by RFBDB-GAN. Training and testing are performed using SOTA methods and the proposed LWDNet.

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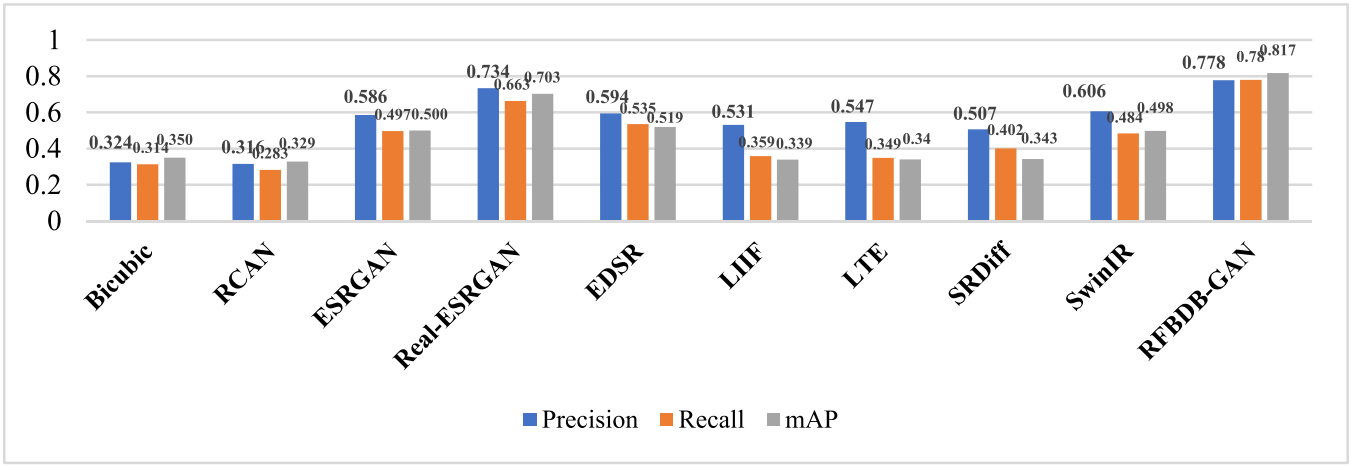


Fig. 17. Results from the LWDNet network for the detection of TLB from SR images generated by different SR methods.

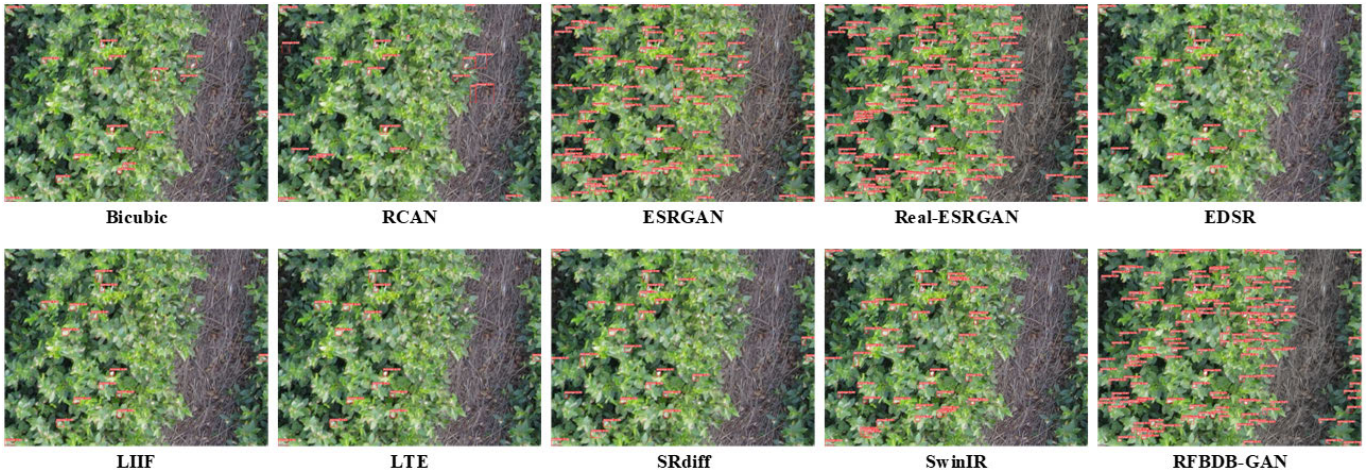


Fig. 18. TLB detection results using LWDNet for SR images generated by different methods.



Fig. 19. Comparison of TLB detection results from different detection networks on SR images generated by RFBDB-GAN.

The detection results from the different detection networks on the same test set are compared in Table VI and Fig. 19. The size of the LWDNet in this study is only 0.7 MB, which is much smaller than some existing deep networks. In addition, the detection accuracy of LWDNet is even higher than that of the SOTA lightweight network YOLOv7-tiny. The TLB spots are small objects with complex backgrounds. Faster R-CNN, SSD YOLOv4, DiffusionDet, and Deformable-DETR are not suitable for this task. The designed LWDNet is lightweight for real-time tea monitoring using a UAV, ensuring accuracy with SIoU loss. The size of the LWDNet is sharply reduced to 716 kB, while its accuracy is only 0.01 less than the baseline network YOLOv5s.

From Table VI and Fig. 19, it can be seen that Deformable-DETR, DiffusionDet, YOLOv4, and so on are significantly inferior to YOLOv5 and YOLOv7 in terms of detection accuracy. YOLOv5L has the highest accuracy and mAP values, but YOLOv5L has many more network parameters than LWDNet. Compared to YOLOv5L, there are more missed and false detections in the LWDNet's detection results, and this will form a focus for improvement in future work. However, the parameters and accuracy

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TABLE VI
TLB DETECTION RESULTS FROM DIFFERENT DETECTION NETWORKS

Network	Precision (%)	Recall (%)	mAP (%)	FLOPs(G)	Params (MB)	Size(MB)
YOLOv5s	78.1	80.3	82.5	7.90	7.01	14.12
YOLOv5L	81.4	79.2	85.4	53.89	46.11	92.80
YOLOv5M	77.0	78.8	82.4	20.85	20.85	42.22
SSD	67.4	76.8	41.5	136.92	23.75	95.03
Faster R-CNN	58.1	61.0	38.7	473.24	28.28	113.44
YOLOv4-tiny	79.2	45.8	59.3	8.07	5.87	23.63
YOLOv4-large	81.3	41.7	63.0	70.72	63.94	256.32
YOLOv7-tiny	76.1	78.5	80.3	6.55	6.01	12.31
DiffusionDet	48.6	43.2	58.8	-	-	1331.2
Deformable-DETR	45.3	37.0	55.8	195.93	39.82	478.6
LWDNet	77.8	78.0	81.7	0.36	0.21	0.72

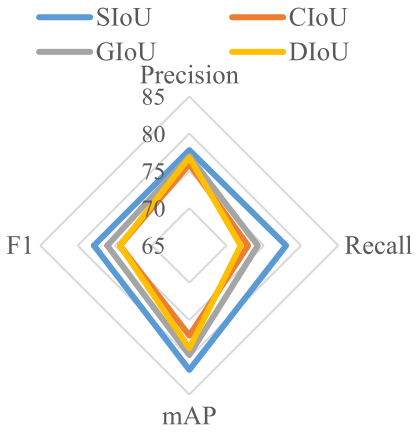


Fig. 20. Comparison of detection results using different loss functions.

of the LWDNet proposed in this study are superior compared with existing lightweight networks such as YOLOv7-tiny.

3) Ablation Experiments on LWDNet: To evaluate the effectiveness and feasibility of the proposed detection network, ablation experiments were conducted to verify the performance of the different components of the LWDNet. These consisted of four sets of experiments involving different loss functions, with and without data augmentation, different feature extraction structures, and different feature fusion structures.

a) Experimental results for different loss functions: The SIoU loss is used in this study to improve the detection accuracy. Fig. 20 shows a comparison of the detection results using different loss functions. Overall, the SIoU loss gives the most obvious improvement in precision, recall, mAP, and F1 for LWDNet.

b) Experimental results for data augmentation: The training dataset was obtained by cropping UAV images of tea plants taken at a low altitude of 2–4 m, and 996 training data samples were obtained after selecting. Since the accuracy of the detection network is closely related to the size of the training set, mosaic enhancement, rotation, and noise addition were applied to expand the dataset, finally giving 3930 samples for training. Table VII shows the experimental results for data augmentation, and it can be seen that data augmentation of the training set was beneficial in improving the detection accuracy of the network.

TABLE VII
RESULTS OF DATA AUGMENTATION

Data Augmentation	Precision (%)	Recall (%)	mAP (%)	F1
✓	77.8	78.0	81.7	77.8
—	75.8	74.7	78.1	75.2

TABLE VIII
EXPERIMENTAL RESULTS FROM DIFFERENT FEATURE EXTRACTION STRUCTURES

Feature Extraction	Precision (%)	Recall (%)	mAP (%)	Size (MB)
Baseline	78.1	80.3	82.5	14.1
C3CBAM	80.6	78.4	82.7	42.7
MobileNetV3	76.0	75.4	76.2	7.5
ShuffleNetV2	71.2	70.8	74.3	1.1
SNet	78.7	75.2	81.1	1.2

c) Experimental results for different feature extraction structures: The proposed LWDNet network adopts the SNet structure for feature extraction, and the same datasets were used for the ablation experiments with different feature extraction structures. The loss function considered in each case was the SIoU function, and the feature fusion part of the detection network remained the same as the baseline network YOLOv5s. Only the feature extraction structure of the detection network was changed. Table VIII shows the detection results using different structures of the feature extraction, and it can be seen that the ShuffleNetV2 structure significantly reduces the number of parameters for the detection network. However, this structure can lead to excessive reduction in accuracy, and the SNet structure was chosen instead for feature extraction, as it is an improved version of the ShuffleNetV2 structure. Our experimental results show that using SNet can reduce the number of network parameters and keep the accuracy fluctuations small.

d) Experimental results from different feature fusion structures: In our framework, the ELPNet structure was used

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TABLE IX
EXPERIMENTAL RESULTS FOR DIFFERENT FEATURE FUSION STRUCTURES

Network	Precision (%)	Recall (%)	mAP (%)	Size (MB)
YOLOv5s	78.1	80.3	82.5	14.1
YOLOv5s+ELPNet	81.3	77.4	81.9	8.7
YOLOv5s-SNet	77.7	75.2	81.1	1.2
LWDNet	77.8	78.0	81.7	0.7

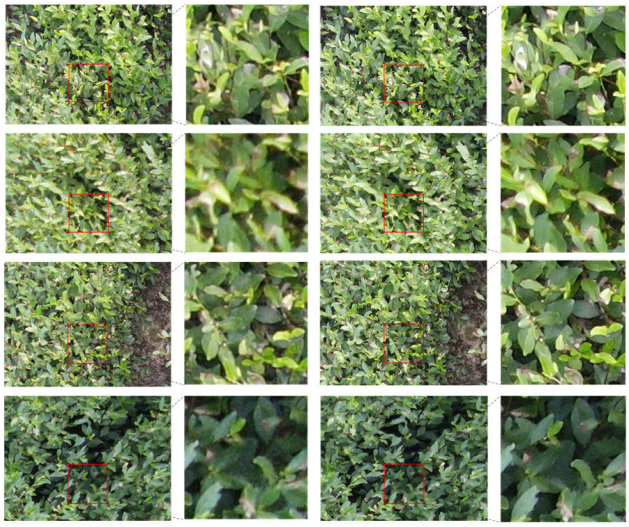
for feature fusion. ELPNet is an efficient lightweight structure that can reduce the parameters of the network without affecting its accuracy. The same dataset was used for the ablation experiments with different feature fusion structures, and the loss function was SIoU in each case.

Table IX shows the experimental results from different feature fusion structures. In the table, the notation YOLOv5s + ELPNet indicates that the ELPNet structure is used to replace the feature fusion structure of the baseline network, and YOLOv5s-SNet indicates that SNet is adopted as the feature extraction part of the network, where the feature fusion part remains the same as the baseline network. The LWDNet structure means that on the basis of YOLOv5s-SNet, the ELPNet structure is adopted as the feature fusion part of the network. Our experimental results show that the ELPNet structure proposed in this study can significantly reduce the number of network parameters, especially when it is combined with the SNet structure and the SIoU loss function, which can not only reduce the network parameters but also improve the detection accuracy of the network.

E. Discussion

In this study, we propose an efficient, low-cost method of detecting TLB disease form LR UAV images of tea plants. The proposed framework combines an RFBDB-GAN SR network with an LWDNet detection network to provide a two-stage detection method, to improve the detection accuracy of TLB. The results of the detection experiments show that the proposed method can improve the detection accuracy of TLB from LR UAV images.

1) Previous studies have relied on mobile phones or DSLRs to capture HR images of tea plants at close range, and data collection via UAV is uncommon for detection applications involving tiny, crowded objects such as areas of tea leaf disease. The strategy presented in this study entailed flying a UAV at a height of 12–14 m and taking LR images for TLB detection. Given the negative impact of a highflying altitude on TLB detection via UAV, we employ the RFBDB-GAN network to perform SR on LR tea images and use a multiscale two-stage degradation method to manually construct the training data for the RFBDB-GAN network. SN terms are added to the discriminator to increase the stability of the network, and RFB dense residual structures are added to the generator to increase the capacity of the network to recover image features. In the LWDNet detection network, the ELPNet structure is designed to carry out feature fusion, which in conjunction with the SIoU loss function can ensure the accuracy of the network while reducing the number of parameters. Our experimental results show that the proposed method significantly improves the effect of TLB detection from LR images of tea plants.
2) TLB spots occupy very few pixels in the UAV images taken at 12–14 m, meaning that the edge and texture features are not obvious or even missing. Most state-of-the-art object detection methods are not suitable for use with LR UAV images. We therefore developed RFBDB-GAN to improve the resolution of these images. We analyze the impact of SR on the detection accuracy from the following three perspectives.
   a) *Reduction of Blurring and Confusion:* LR images may be affected by blurring or confusion, which may lead to inaccurate object detection. The use of HR images can reduce this phenomenon and improve the detection accuracy.
   b) *More Detailed Information:* HR images contain more detailed information, meaning that the edges, textures, shapes, and other features of the object are more clearly visible, which helps the model detect the object more accurately.
   c) *Extraction of More Distinguishable Features:* More key points, edges, and structures are available for feature extraction in HR images; this enables image processing and object detection models to more fully extract distinguishable features to detect objects.
3) Due to a lack of matching pairs of ground-truth and LR images acquired at a height of 12-14 m for SR experiments, merely applying SR evaluation metrics to explain the restoration performance of the network is hardly compelling. Fig. 21 shows a comparison between images taken at a height of 7.5 m and the SR images generated by RFBDB-GAN. Visual experts were invited to observe and compare these images, and the results of



(a) (b)

Fig. 21. Comparison of tea images before and after SR at a shooting height of 7.5 m. (a) Original images. (b) SR images.

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Fig. 22. Detection results from LWDNet when applied to the original images and SR images generated by different methods. (a) Original. (b) RCAN. (c) ESRGAN. (d) Real-ESRGAN. (e) RFBDB-GAN.

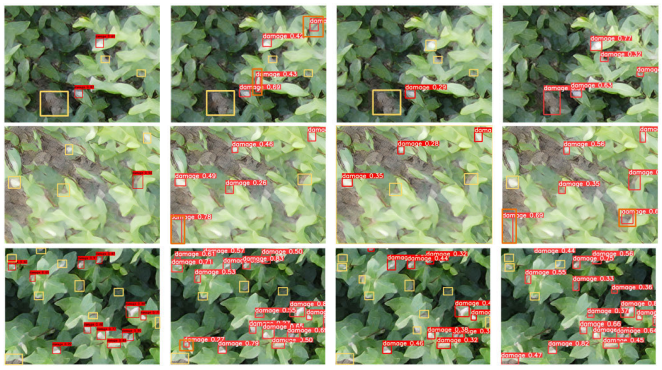


Fig. 23. Detection results from different detection networks when applied to SR images generated by RFBDB-GAN. (a) YOLOv4-tiny. (b) YOLOv5s. (c) YOLOv7-tiny. (d) LWDNet.

this analysis showed that the SR images were basically consistent with the original images and the details of the tea leaves in the SR images were clearer, resulting in a better visual effect. Although RFBDB-GAN can improve the details of the tea images, this network has a large number of parameters, and the training procedure is time-consuming and complex, meaning that it is far from the lightweight network envisaged. Subsequent research on the detection of tea diseases will concentrate on lighter and more efficient SR methods. In this study, we conducted detection experiments for TLB from the SR images.

4) The detection results are shown in Figs. 18 and 19, and the resolution of the 4× SR images is 3200 × 2400. To simplify further investigation of the detection results, each SR image was cropped to give 16 images with a resolution of 800 × 600. Fig. 22 shows the detection results from LWDNet when applied to the original images and the SR images generated by different SR methods (note that we mainly compare RFBDB-GAN with GAN-based methods). The yellow boxes in the images indicate the missed detections, the orange boxes indicate the false detections, and the red boxes show the detection results from the net. The areas occupied by TLB spots are very small and the information on the spots is not abundant, meaning that it is easy to misdetections. RFBDB-GAN can effectively improve the resolution of the images, and the missed detections are therefore significantly reduced. From a comparison of the last rows of Fig. 22(d) and (e), it can be seen that the Real-ESRGAN network overfits the dead leaves on the ground, causing the network to misdetect the background as the TLB.
5) The proposed LWDNet is a lightweight network, whereas most of the networks listed in Table VI have large numbers of parameters. Since SSD, Faster R-CNN, DiffusionDet, and Deformable-DETR do not give significant improvements, only LWDNet and the YOLO-tiny/s series are discussed here for brevity. The test images were resolved by RFBDB-GAN and cropped to give images with a resolution of 800 × 600. The detection results from YOLOv4-tiny, YOLOv5s, YOLOv7-tiny, and LWDNet are shown in Fig. 23. Each detection network has both missed detections and false detections for TLB. When detecting small TLB spots, LWDNet also focuses on the larger spots, whereas other networks, such as YOLOv5s, ignore the larger spots to a certain extent. In addition, the results reveal that the LWDNet network tends to give missed detections of TLB at the leaf edges and incorrectly detects the ground as TLB. Subsequent work will focus on reducing the false and missed detections of LWDNet to further increase the detection accuracy.
6) Bao et al. [63] previously explored a TLB detection approach based on a combination of RCAN and DDMA-YOLOv5, which had a detection accuracy of 73.3%. They captured images of tea plants with a UAV flying at a height of around 5 m, and 2× SR recovery of images was performed using the RCAN network. Detection of TLB was carried out by the DDMA-YOLOv5. Although the detection accuracy of their DDMA-YOLOv5 network was 3.8% higher than that of YOLOv5s, DDMA-YOLOv5 added 2-D attention modules and RFB structures based on YOLOv5s, which caused the number of parameters to be much higher than baseline. In this experiment, 2× SR was performed on images of tea by RFBDB-GAN, with the same annotation method as used by Bao et al. [63] (diseased leaves were labeled rather than just TLB spots), and LWDNet was used for detection. Table X shows the results of this experiment.
7) SuperYOLO [64] is a multitask learning framework. We conducted experiments using SuperYOLO on our TLB dataset, and the quantitative results of the experiments are shown in Table XI. It can be seen that the detection results on TLB for LR tea plant images using SuperYOLO alone were not satisfactory. However, when RFBDB-GAN was combined with SuperYOLO, all of the quantitative indices showed improvement, and the mAP index in particular increased by 34%. SuperYOLO utilizes EDSR as an SR branch. When both RFBDB-GAN and EDSR are applied to the data simultaneously, EDSR may cause the fine details already restored by RFBDB-GAN to become smoothed

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TABLE X
COMPARISON OF DETECTION RESULTS FROM THE PROPOSED METHOD AND THAT OF BAO ET AL.

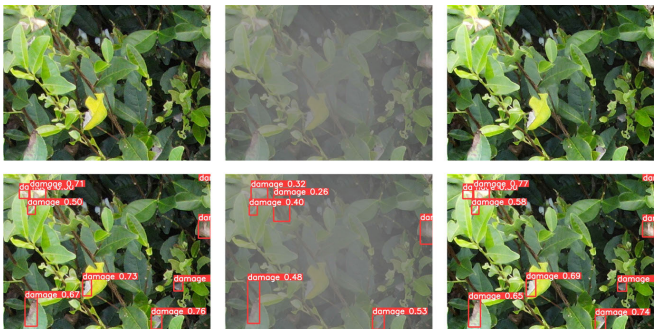
Method	Precision (%)	Recall (%)	F1	AP@0.5 (%)	AP@0.5:0.95 (%)	Size (MB)
Bao et al.	72.5	65.3	68.7	73.3	42.6	>14.4
Our method	73.2	69.6	71.4	74.7	44.2	0.7

TABLE XI
COMPARISON OF DETECTION RESULTS FROM THE PROPOSED METHOD AND THAT OF SUPERYOLO

RFBDB-GAN	SuperYOLO	LWDNet	Precision (%)	Recall (%)	F1	mAP@50 (%)
—	✓	—	58.4	39.5	47.1	40.6
—	—	✓	54.3	38.7	45.1	34.5
✓	✓	—	77.7	70.9	74.1	74.1
✓	—	✓	77.8	78.0	77.9	81.7

TABLE XII
EXPERIMENTAL RESULTS UNDER UNEVEN HAZING CONDITIONS

	Precision (%)	Recall (%)	mAP@0.5 (%)	AP@0.5:95 (%)	F1
Original	78.9	74.2	81.7	36.8	76.4
Uneven light hazing	74.9	54.3	64.4	27.4	62.9
Defogging	75.3	72.5	74.9	33.7	73.8



(a) (b) (c)

Fig. 24. Experimental results of image dehazing and TLB detection. (a) Original image. (b) Uneven light hazing image. (c) RFBDB-GAN defogging image.

out, meaning that a combination of RFBDB-GAN and LWDNet produces better results than a combination of RFBDB-GAN and SuperYOLO. Hence, to prevent conflicts in data restoration caused by different SR models, the use of multiple SR models simultaneously should be avoided.

8) We applied artificial uneven light hazing [65] to our TLB data. The proposed two-stage method for dehazing and TLB detection was applied, and the experimental results are shown in Table XII and Fig. 24. When detecting TLB from the hazing images, the model had a high ratio of false negatives, and the accuracy was not ideal. However, after applying RFDB-GAN to dehaze the images, there was a significant improvement in the detection indices, and in particular, the mAP was increased by 10%. We note that our study was primarily focused on TLB data characterized by uneven lighting and LR. The RFBDB-GAN model was specifically designed to address the degradation caused by the different shooting heights of the UAV. Although the proposed model was not designed for image dehazing, it was able to achieve good performance on images with artificially uneven light hazing.

V. CONCLUSION

Areas of TLB are often blurred and difficult to detect from LR UAV images of tea plants, and the backgrounds are complex, making it challenging to achieve accurate detection using existing TLB detection methods. This study has therefore put forward an effective and affordable approach for detecting TLB from LR UAV images. Our framework combined the proposed RFBDB-GAN and LWDNet models, giving a two-stage method, in order to increase the detection accuracy of TLB. RFBDB-GAN adopts the RFBDB structure, which improves the network's ability to recover the details of tea leaves. In order to train the RFBDB-GAN network effectively, a multiscale two-stage degradation method was developed to construct pairs of training images, and our experimental results verified the effectiveness of this method for the recovery of LR tea images. LWDNet uses ELPNet and SNet structures as the network's feature fusion and feature extraction parts, respectively. The parameters of the baseline network were reduced, with the size of the LWDNet model being only 0.7 MB and the number of parameters only 0.2 MB. The results of a detection experiment showed that the proposed method could improve the detection accuracy of TLB from LR UAV images compared to existing methods.

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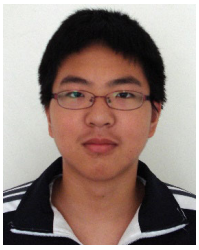
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